First Direct Measurement of Cosmic Circumference: 379 Gpc if SKA-Low detects 472,789,997.636 Hz

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Abstract

If CHIME (2023), DESI (2024), and Pantheon+ (2022) measurements are all correct, the universe must be closed with circumference $C = 379 \pm 38$ Gpc. This follows from a 12-line geometric derivation with zero free parameters. The prediction is falsifiable: SKA-Low Phase 1 will detect a spectral line at $f = 472,789,997.636$ Hz with 144σ significance in 400h integration, or the derivation is wrong. Code: `https://github.com/francodealva/Cosmiccone`

1 The Tension Triplet

Three independent probes give:

$$H_0^{\text{CHIME}} = 74.4 \pm 1.0 \text{ km/s/Mpc} \tag{1}$$

$$H_0^{\text{Pantheon+}} = 73.4 \pm 1.0 \text{ km/s/Mpc} \tag{2}$$

$$w_{\text{DESI}} = -0.65 \pm 0.05 \tag{3}$$

The $\sim$2% H_0 difference is within errors. The critical input is $w < -1/3$.

2 Geometric Closure

For $w < -1/3$, the Friedmann equation requires positive curvature. The curvature radius is:

$$R = \frac{c}{H_0}\sqrt{\frac{-3w}{1+w}} = 60.3 \text{ Gpc} \tag{4}$$

using $H_0 = 73.9$ km/s/Mpc average. The circumference follows:

$$C = 2\pi R = 379 \text{ Gpc} \tag{5}$$

3 Falsifiable Prediction

A closed universe has a fundamental mode at wavelength $\lambda = C$:

$$f = \frac{c}{C} = \frac{299792458 \text{ m/s}}{379 \text{ Gpc}} = 472,789,997.636 \text{ Hz} \tag{6}$$

Test: SKA-Low Phase 1, 400h integration at 472.790 MHz.
Signal: $> 144\sigma$ if $C = 379$ Gpc.
Null: No detection rules out this geometry.

4 Conclusion

No free parameters were fit. If SKA-Low detects $f = 472,789,997.636$ Hz, we have measured the size of the universe. If not, the assumption that all three datasets are correct fails.

Repository: `https://github.com/francodealva/Cosmiccone`